

Simulating Dark Matter and Gas in the Bullet Cluster

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ABSTRACT

We present idealized N -body/SPH simulations of the Bullet Cluster merger using recent cluster masses inferred from gravitational-lensing observations. Spherical cluster models were first used to study the effects of the initial velocity, impact parameter, and gas core radii. The preferred model used an initial velocity of 1000 km s^{-1} , an impact parameter of 300 kpc, a main-cluster gas core radius of $r_c = r_s/2$, and a bullet-cluster gas core radius of $r_c = r_s/5$. Triaxial dark matter halos were then introduced to better reproduce the elongated mass distributions seen in recent lensing maps. The simulations were compared shortly after the first core passage, at a projected dark matter peak separation of approximately 750 kpc. Synthetic Chandra observations were produced to compare the simulated gas distribution with the observed X-ray morphology. The preferred spherical model produced a clear separation between the gas and dark matter components and reproduced the broad wedge-shaped morphology of the X-ray-emitting gas. The triaxial models improved the shape and orientation of the projected dark matter contours. However, the detailed X-ray morphology and observed shock structure were not fully reproduced, indicating that additional gas physics and a more systematic parameter search may be required.

Keywords: galaxy clusters — intracluster medium — hydrodynamical simulations — dark matter

1. INTRODUCTION

Galaxy clusters are the largest gravitationally bound structures in the Universe, with typical masses of 10^{14} – $10^{15} M_\odot$. Approximately 80% of their mass is dark matter, while the remaining mass is baryonic. Most of the baryonic component is contained in the hot intracluster medium (ICM), with only a small fraction contained in galaxies. During cluster mergers, the high collision velocities and large masses involved make these systems useful laboratories for studying dark matter, the ICM, shocks, and structure formation. These mergers are generally classified according to their mass ratio: major mergers involve clusters of comparable mass, while minor mergers have a secondary cluster with less than approximately 30–40% of the primary mass (Valdarnini 2024).

The target of this project is the Bullet Cluster, 1E 0657–56, a merging galaxy-cluster system observed at a redshift of $z = 0.296$. The system consists of a massive main cluster and a smaller secondary cluster, referred to as the bullet. Chandra observations revealed a prominent bow shock and a cool, dense gas core moving through the ICM of the main cluster (Markevitch et al. 2002, 2004). Gravitational-lensing studies later showed a clear spatial offset between the projected mass centroids and the X-ray-emitting gas (Clowe et al. 2004, 2006). The lensing mass peaks approximately

coincide with the brightest cluster galaxies, while the collisional gas is displaced by ram pressure. Because the ICM contains most of the baryonic mass and the galaxies alone cannot account for the lensing signal, this separation provides direct evidence for a dominant collisionless dark matter component (Clowe et al. 2006).

Recent JWST observations provide updated constraints on the mass and projected shape of the Bullet Cluster. The system consists of a massive main cluster and a smaller secondary cluster, referred to as the bullet. NIRCам imaging and NIRSpectroscopy have produced a more detailed lensing reconstruction of the central mass distribution, showing that the projected mass components are elongated rather than circular (Rihtarišič et al. 2026). A joint JWST–DECam lensing analysis measured masses of approximately $1.51 \times 10^{15} M_\odot$ and $1.49 \times 10^{14} M_\odot$ for the main cluster and the bullet, respectively. The resulting mass ratio of approximately 10 : 1 classifies the system as a minor merger (Cho et al. 2025). The elliptical appearance of the lensing mass distributions also motivates the use of triaxial, rather than spherical, dark matter halos in this work.

Several numerical studies have attempted to reproduce the Bullet Cluster. Springel & Farrar (2007) studied the relation between the bow-shock velocity and the velocity of the bullet subcluster. Mastropietro & Burkert (2008) used idealized N -body/SPH simulations to explore the effects of mass ratio, impact parameter, concentration, and initial velocity.

* Research supervised by Dr. Joseph MacMillan



Figure 1. Composite observation of the Bullet Cluster. Pink traces the X-ray-emitting intracluster gas, while blue shows the projected mass distribution inferred from gravitational lensing. The visible galaxies are shown in the optical and near-infrared background. Image credit: NASA, ESA, CSA, STScI, and CXC.

Lage & Farrar (2014) later used triaxial initial conditions and magnetohydrodynamical simulations constrained by gravitational-lensing and Chandra observations. Together, these studies showed that the observed morphology depends strongly on both the merger parameters and the internal structure of the initial clusters.

In this project, we use idealized N -body/SPH simulations based on the most recent mass estimates and projected dark matter maps of the Bullet Cluster (Rihtaršič et al. 2026; Cho et al. 2025). Most of the parameter exploration was performed using spherical cluster models, while triaxial models were introduced during the final stage to study the elongated lensing distribution. We focus on the offset between the dark matter and ICM components and on the morphology of the X-ray-emitting gas. Synthetic Chandra maps are produced for the preferred simulations and compared with the observed X-ray morphology.

Section 2 describes the spherical cluster models, their initial conditions, equilibrium tests, and collision simulations. Section 3 presents the generation and simulation of the triaxial models. Section 4 compares the spherical and triaxial results with the observed dark matter and X-ray morphology. The main conclusions and possible extensions of this work are presented in Section 5.

2. SPHERICAL CLUSTERS

The initial conditions were developed in three stages. At the beginning of the project, we compared CLUSTEP (Ruggerio & Lima Neto 2017) with the Cluster Generator library developed by John ZuHone (ZuHone 2024). Cluster Generator was then used to generate the spherical cluster models for the main parameter exploration. During the final stage of the project, triaxial dark matter and gas models were generated

to better reproduce the elongated mass distributions seen in the lensing maps.

2.1. Cluster Parameters

The main and bullet cluster masses were taken from the recent joint JWST-DECam lensing analysis of Cho et al. (2025). We adopted $M_{200} = 1.511 \times 10^{15} M_{\odot}$ for the main cluster and $M_{200} = 1.49 \times 10^{14} M_{\odot}$ for the bullet cluster. Both clusters were modeled at the observed redshift $z = 0.296$, using $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_M = 0.3$, and $\Omega_{\Lambda} = 0.7$.

The cluster mass is defined as the mass M_{200} enclosed within the radius R_{200} . This radius is defined such that the mean enclosed density is 200 times the cosmological critical density at the cluster redshift. Therefore,

$$M_{200} = \frac{4\pi}{3} (200) \rho_{\text{crit}}(z) R_{200}^3. \quad (1)$$

From the Friedmann equation, the critical density at the cluster redshift was calculated as

$$\rho_{\text{crit}}(z) = \frac{3H^2(z)}{8\pi G}, \quad (2)$$

where G is the gravitational constant and $H(z)$ is the Hubble parameter at redshift z , given by

$$H(z) = H_0 \sqrt{\Omega_M(1+z)^3 + \Omega_{\Lambda}}. \quad (3)$$

Solving Equation (1) for R_{200} gives

$$R_{200} = \left[\frac{3M_{200}}{4\pi(200)\rho_{\text{crit}}(z)} \right]^{1/3}. \quad (4)$$

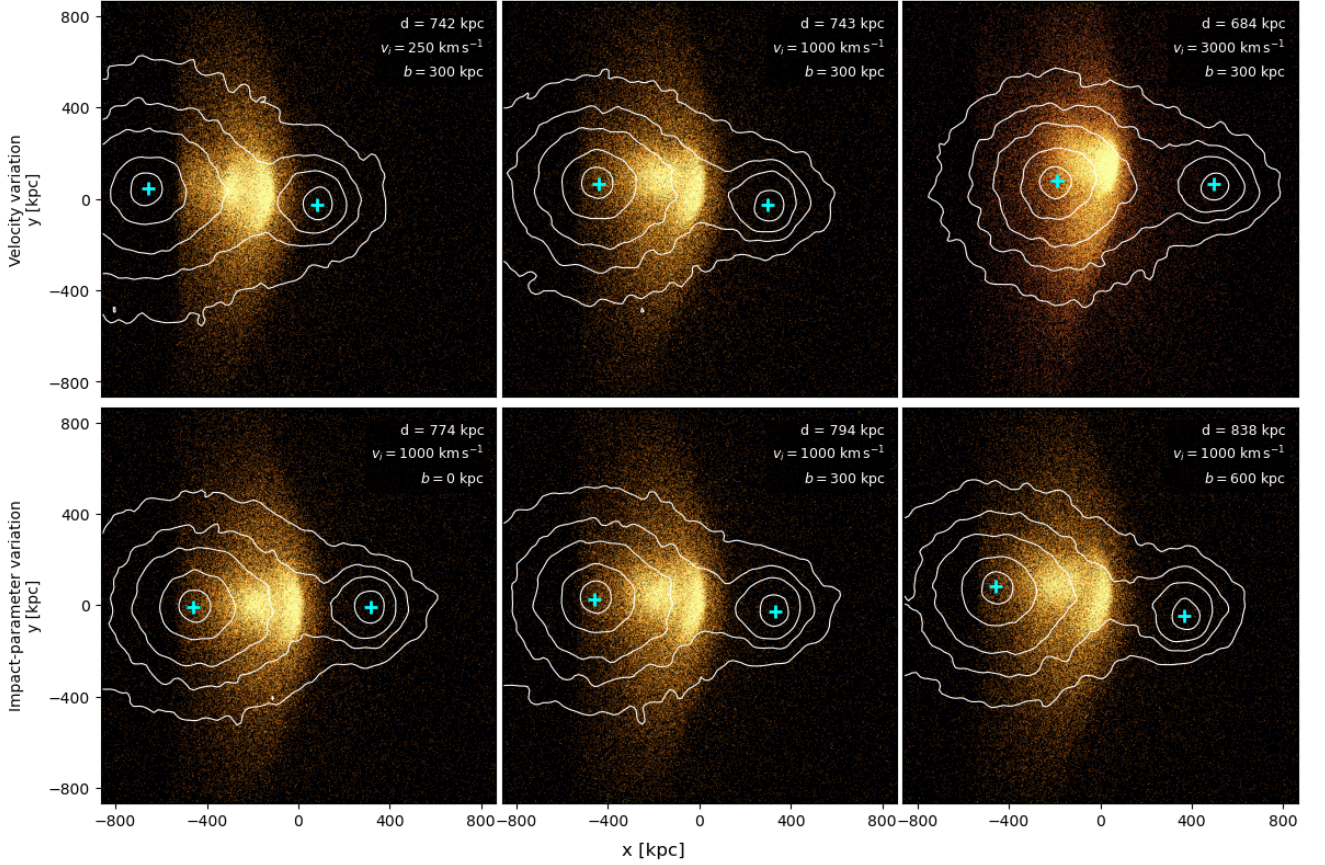


Figure 2. Representative spherical collision simulations. The orange maps show the mock Chandra X-ray emission, the white contours show the projected dark matter distribution, and the cyan crosses mark the dark matter density peaks. The upper row varies the initial velocity while keeping $b = 300$ kpc, whereas the lower row varies the impact parameter while keeping $v_i = 1000$ km s⁻¹. The projected dark matter peak separation d is shown in each panel.

The dark matter distribution of each cluster was modeled using a Navarro–Frenk–White (NFW) profile (Navarro et al. 1997),

$$\rho_{\text{DM}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}, \quad (5)$$

where ρ_s is the characteristic density and r_s is the scale radius. The concentration parameters were estimated using the mass–concentration relation of Duffy et al. (2008),

$$c_{200} = 5.71 \left(\frac{M_{200}}{2 \times 10^{12} M_{\odot} h^{-1}} \right)^{-0.084} (1 + z)^{-0.47}, \quad (6)$$

where $h = H_0/(100 \text{ km s}^{-1} \text{ Mpc}^{-1}) = 0.7$. The scale radius was then defined as

$$r_s = \frac{R_{200}}{c_{200}}. \quad (7)$$

The resulting cluster parameters are listed in Table 1.

2.2. Initial Conditions

Table 1. Adopted masses and derived structural parameters of the main and bullet clusters.

Cluster	M_{200} [$M_{\odot}$]	R_{200} [kpc]	c_{200}	r_s [kpc]
Main	1.511×10^{15}	2140.13	2.99	715.76
Bullet	1.49×10^{14}	988.74	3.63	272.38

2.2.1. Dark Matter

Cluster Generator (ZuHone 2024) was used to create the spherical initial conditions. The dark matter followed the NFW profile defined in Equation (5), using the parameters listed in Table 1.

To limit the initial extent of each dark matter halo, an exponential cutoff was applied beyond R_{200} . Following Valdarnini (2024), the density outside R_{200} was written as

$$\rho_{\text{DM}}(r) = \rho_{\text{DM}}(R_{200}) \left(\frac{r}{R_{200}} \right)^{\delta} \exp \left[-\frac{r - R_{200}}{r_{\text{decay}}} \right], \quad (8)$$

where

$$\delta = -\frac{R_{200}}{r_{\text{decay}}} - \frac{1 + 3c_{200}}{1 + c_{200}}. \quad (9)$$

We used $r_{\text{decay}} = 0.2R_{200}$ and generated the dark matter profile out to $2R_{200}$.

2.2.2. Gas

The gas followed an isothermal beta profile (Cavaliere & Fusco-Femiano 1976),

$$\rho_g(r) = \rho_0 \left[1 + \left(\frac{r}{r_c} \right)^2 \right]^{-3\beta/2}, \quad (10)$$

where ρ_0 is the central gas density, r_c is the gas core radius, and β controls the outer slope of the profile. We used $\beta = 2/3$ and varied the gas core radius to study how the initial gas distribution affected the final X-ray morphology. For both the main and bullet clusters, we generated models with $r_c/r_s = 1/5, 1/4, 1/3, 2/5$, and $1/2$. This produced 25 different combinations of main and bullet gas profiles.

The gas mass inside R_{200} was calculated using

$$M_{\text{gas}}(< R_{200}) = \frac{f_g}{1 - f_g} M_{\text{DM}}(< R_{200}), \quad (11)$$

where f_g is the gas mass fraction within R_{200} , defined as $f_g = M_{\text{gas}}/(M_{\text{gas}} + M_{\text{DM}})$. We adopted $f_g = 0.13$. Cluster Generator then calculated the gas temperature required for hydrostatic equilibrium. These models were used to explore how the gas core radius, initial velocity, and impact parameter affected the merger. The complete simulation grid is presented in Section 2.4.

2.3. Isolated Stability Tests

The spherical main and bullet clusters were evolved in isolation for 2 Gyr before being used in the collision simulations. The radial dark matter and gas density profiles were measured at $t = 0, 1$, and 2 Gyr. Figure 4 shows the profiles as a function of radius, with the dotted lines indicating R_{200} for each cluster.

The dark matter profiles remained nearly unchanged throughout the test. The gas profiles also remained stable within R_{200} , with only small differences in the innermost and outermost radial bins. These tests show that the adopted spherical initial conditions maintained their density structure over the timescale relevant to the collision simulations.

2.4. Collision Simulations

2.4.1. Numerical Setup

The collision simulations were performed using the GADGET-4 code (Springel et al. 2021). Dark matter was represented by collisionless particles, while the intracluster gas was evolved using smoothed-particle hydrodynamics (SPH). The gas was treated adiabatically, without radiative

Table 2. Masses and particle numbers used in the spherical collision simulations.

Cluster	$M_{200} [M_\odot]$	N_{DM}	N_{gas}
Main	1.511×10^{15}	8.0×10^5	1.0×10^6
Bullet	1.49×10^{14}	8.0×10^4	1.0×10^5
Total	—	8.8×10^5	1.1×10^6

cooling, star formation, feedback, or magnetic fields. A gravitational softening length of approximately 5.4 kpc was used. The mass and particle resolution of each cluster are listed in Table 2.

The clusters were initially separated by 7000 kpc. This distance is larger than twice the sum of their R_{200} radii, reducing the initial overlap between their extended dark matter halos and allowing the clusters to approach before the main interaction begins. This separation was kept fixed throughout the spherical parameter exploration.

2.4.2. Initial Velocity and Impact Parameter

We first studied the effects of the initial velocity and impact parameter on the collision. Several dozen simulations were performed using initial velocities between 250 and 3000 km s^{-1} and impact parameters between 0 and 600 kpc. The impact parameter describes how far the initial trajectory is from a head-on collision.

The initial velocity affected the time of core passage, the strength of the ram pressure, and the separation produced between the gas and dark matter. The impact parameter affected the closest approach of the cluster centers and the asymmetry of the resulting gas distribution. As also found by Mastropietro & Burkert (2008), head-on collisions produced a gas morphology that was too symmetric, while an offset collision produced a more asymmetric main cluster and increased the removal of gas from the bullet.

The preferred combination was an initial velocity of 1000 km s^{-1} and an impact parameter of 300 kpc. This model produced a clear separation between the gas and dark matter while maintaining an X-ray morphology similar to the observed system. Figure 2 shows representative simulations from this parameter search. All simulations are shown shortly after the first core passage, using the snapshot with a projected dark matter peak separation closest to 750 kpc, approximately matching the separation observed on the sky.

2.4.3. Gas Core Radius

The initial gas distribution also has an important effect on the final X-ray morphology. Mastropietro & Burkert (2008) adopted a gas core radius of $r_c = r_s/2$, while Valdarnini (2024) showed that changing the core radius can substantially modify the gas distribution produced by a cluster collision. Lage & Farrar (2014) similarly found that reproducing the observed X-ray emission depends strongly on the initial gas profiles.

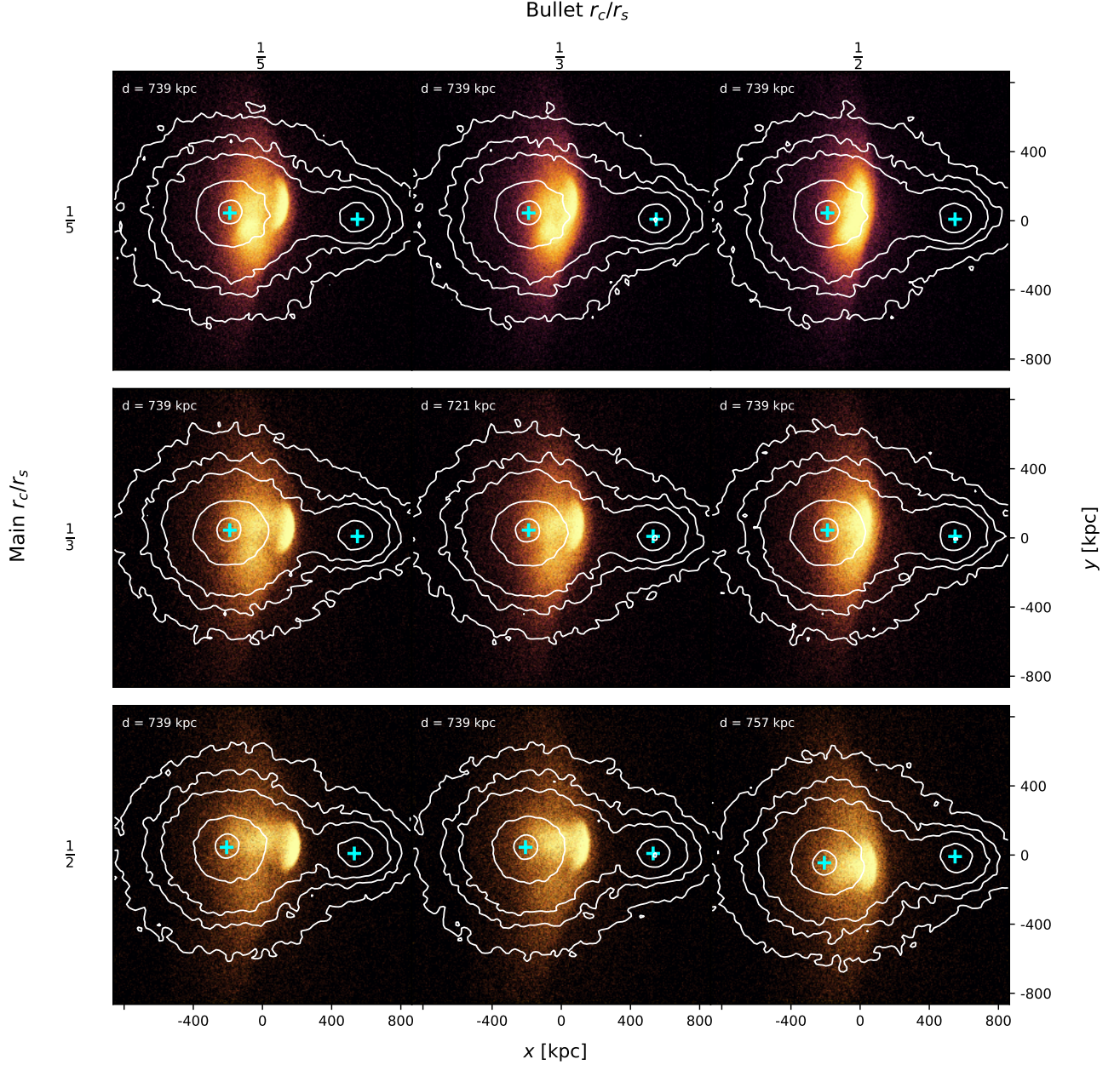


Figure 3. Representative simulations from the gas core-radius search. Rows correspond to the main-cluster value of r_c/r_s , while columns correspond to the bullet-cluster value. All simulations use $v_i = 1000 \text{ km s}^{-1}$ and $b = 300 \text{ kpc}$. Orange shows the mock Chandra X-ray emission, white contours show the projected dark matter distribution, and cyan crosses mark the dark matter density peaks. The projected peak separation d is shown in each panel.

For each cluster, we tested gas core radii of $r_c/r_s = 1/5$, $1/4$, $1/3$, $2/5$, and $1/2$. The main- and bullet-cluster core radii were varied independently, producing 25 simulations. All models used the preferred initial velocity of 1000 km s^{-1} and impact parameter of 300 kpc .

Figure 3 shows nine representative combinations from the complete 5×5 search. The rows vary the main-cluster core radius, while the columns vary the bullet-cluster core radius. All simulations are shown shortly after the first core passage, using the snapshot with a projected dark matter peak separation closest to 750 kpc . This approximately matches the separation observed on the sky and allows the gas morphologies to be compared at a similar stage of the collision.

3. TRIAXIAL CLUSTERS

3.1. Dark Matter

For the final part of the project, we attempted to recreate the best-fitting initial conditions of [Lage & Farrar \(2014\)](#). Their model used triaxial dark matter and gas distributions

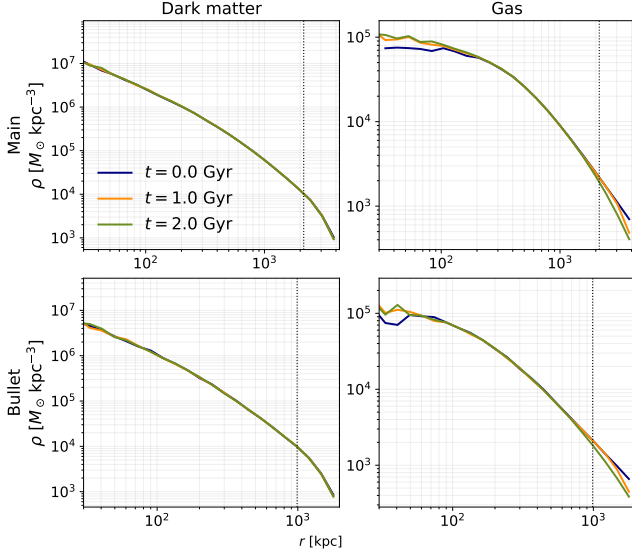


Figure 4. Radial dark matter and gas density profiles of the spherical main and bullet clusters evolved in isolation. The profiles are shown at $t = 0, 1$, and 2 Gyr. The dotted lines indicate R_{200} for each cluster.

to reproduce the shapes seen in the gravitational lensing and Chandra observations. Their best-fitting axis ratios were $P = 0.35$ and $Q = 0.68$ for the main cluster, and $P = 0.61$ and $Q = 0.68$ for the bullet cluster. We used these shapes as the starting point for our triaxial models, but replaced their original cluster masses with the recent JWST-DECam estimates used throughout this project. We later generated additional bullet models with $P = 0.45$ and $P = 0.31$ to test whether a more elongated bullet could better reproduce the observed lensing morphology.

The triaxial dark matter halos were generated using the AGAMA Spheroid density model (Vasiliev 2019). Following the notation of Lage & Farrar (2014), the density is written in terms of the ellipsoidal radial parameter R ,

$$R^2 = x^2 + \frac{y^2}{Q^2} + \frac{z^2}{P^2}, \quad (12)$$

where $P = z/x$ and $Q = y/x$ are the minor-to-major and intermediate-to-major axis ratios, respectively. In AGAMA, these correspond to `axisRatioZ = P` and `axisRatioY = Q`.

The general AGAMA Spheroid density profile is

$$\rho(R) = \rho_0 \left(\frac{R}{r_s} \right)^{-\gamma} \left[1 + \left(\frac{R}{r_s} \right)^\alpha \right]^{(\gamma-\beta)/\alpha} \exp \left[- \left(\frac{R}{r_{\text{cut}}} \right)^\xi \right], \quad (13)$$

where α controls the transition between the inner and outer slopes, γ and β are the inner and outer slopes, and ξ controls the strength of the outer cutoff. We used $\alpha = 1$, $\gamma = 1$, and $\beta = 3$, giving the NFW-like profile

$$\rho_{\text{DM}}(R) = \frac{\rho_0}{(R/r_s)(1 + R/r_s)^2} \exp \left[- \left(\frac{R}{r_{\text{cut}}} \right)^\xi \right]. \quad (14)$$

For both clusters, we used $r_{\text{cut}} = 2R_{200}$ and $\xi = 8$.

3.2. Gas

The gas particles were generated using a custom Python code based on the triaxial method of Lage & Farrar (2014).¹ The gas followed the same beta profile used in the spherical models,

$$\rho_g(R) = \rho_0 \left[1 + \left(\frac{R}{r_c} \right)^2 \right]^{-3\beta/2}, \quad (15)$$

where R is the gas radial parameter. We used $\beta = 2/3$, $r_c = r_s/2$ for the main cluster, and $r_c = r_s/5$ for the bullet cluster.

The gas shape was calculated from the equipotential surfaces of the AGAMA dark matter potential. For each value of R , the code first calculated the potential on the major axis, $\Phi(R, 0, 0)$. It then found the positions y_R and z_R with the same potential,

$$\Phi(R, 0, 0) = \Phi(0, y_R, 0) = \Phi(0, 0, z_R). \quad (16)$$

The gas axis ratios were then defined as

$$Q_\Phi(R) = \frac{y_R}{R}, \quad P_\Phi(R) = \frac{z_R}{R}. \quad (17)$$

These axis ratios were calculated between $0.001R_{200}$ and $30R_{200}$. The large radial range was used to avoid extrapolation problems when calculating the outer gas distribution.

The central gas density was chosen so that the gas fraction inside R_{200} was $f_g = 0.13$. The gas mass was normalized using

$$M_{\text{gas}}(< R_{200}) = \frac{f_g}{1 - f_g} M_{200}, \quad (18)$$

After the density was normalized, the gas particles were sampled out to $2R_{200}$. Their positions were assigned using

$$\begin{aligned} x &= R \sin \theta \cos \phi, & y &= Q_\Phi(R) R \sin \theta \sin \phi, \\ z &= P_\Phi(R) R \cos \theta. \end{aligned} \quad (19)$$

This gives the gas distribution the shape of the dark matter equipotential surfaces rather than the more elongated shape of the dark matter density itself.

¹ The source code implementing the triaxial gas generation algorithm described in this section is available at <https://github.com/DiegoRod-13/bullet-cluster-simulations>.

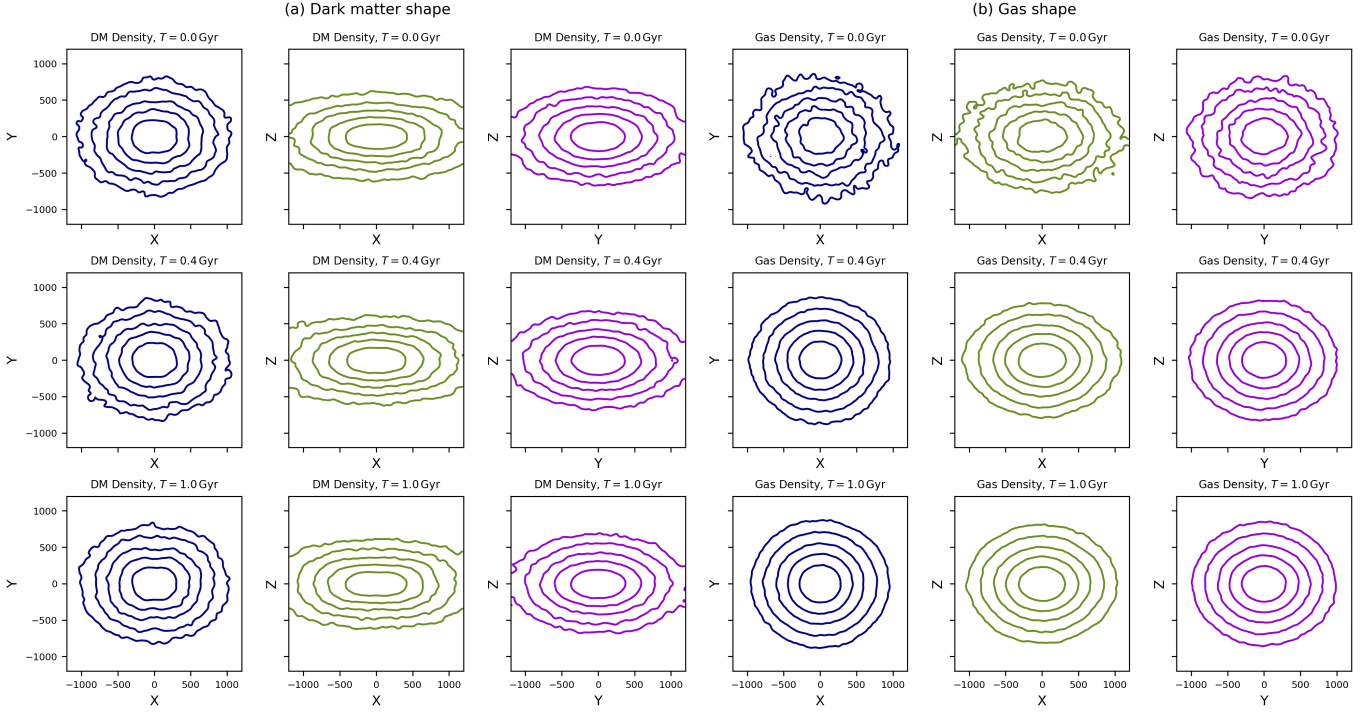


Figure 5. Isolated evolution of a representative triaxial cluster at $t = 0, 0.4$, and 1 Gyr. The dark matter maintains its triaxial shape, while the gas follows the rounder equipotential surfaces.

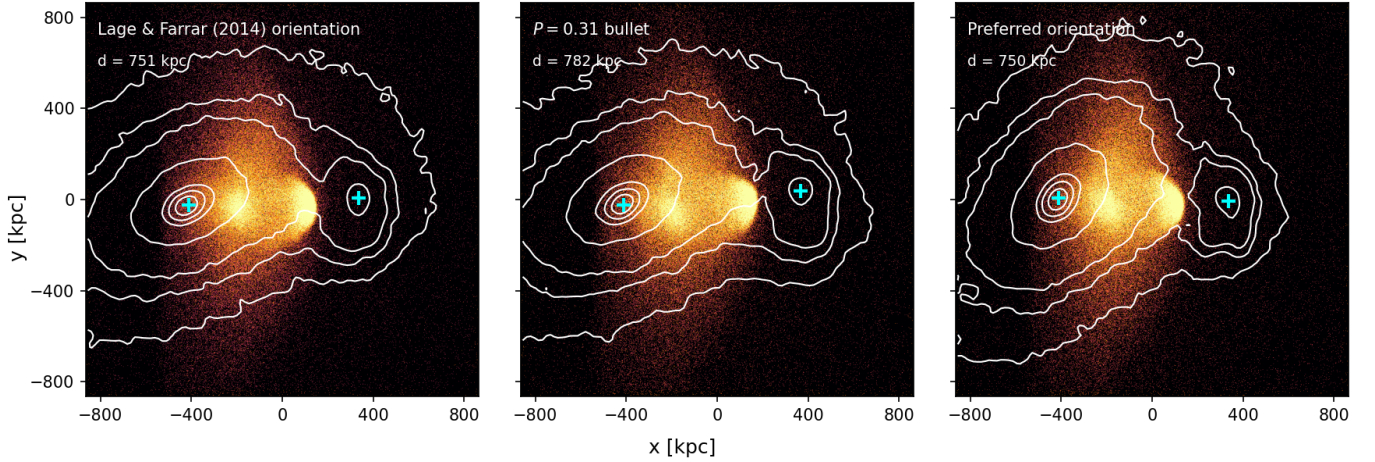


Figure 6. Progression of the triaxial parameter search. The left panel uses the geometry adopted by [Lage & Farrar \(2014\)](#), the middle panel uses the $P = 0.31$ bullet, and the right panel shows the preferred model from the final orientation search. The orange emission shows the mock Chandra X-ray map, the white contours show the projected dark matter distribution, and the cyan crosses mark the dark matter peaks.

The gas internal energy was calculated by assuming hydrostatic equilibrium. The radial potential gradient was measured along the major axis using the AGAMA force,

$$\frac{d\Phi}{dR} = -F_x(R, 0, 0). \quad (20)$$

$$u(R) = \frac{3}{2\rho_g(R)} \int_R^{10R_{200}} \rho_g(R') \frac{d\Phi}{dR'} dR'. \quad (21)$$

The upper integration limit of $10R_{200}$ was used to ensure correct internal energy for particles at large radius. The gas particles were initially assigned zero velocity before the clusters were relaxed in isolation.

The specific internal energy was then calculated as

3.3. Isolated Equilibrium Test

The triaxial cluster was evolved in isolation for 1 Gyr. Figure 5 shows the dark matter and gas density contours along the three principal planes. The dark matter maintained its triaxial shape, while the gas followed the equipotential surfaces, which are rounder than the dark matter density distribution.

3.4. Collision Simulations

The triaxial collision simulations used the preferred spherical collision parameters: an initial separation of 7000 kpc, velocity of 1000 km s^{-1} , and impact parameter of 300 kpc. These parameters were kept fixed while the halo shapes and orientations were varied.

3.4.1. Shape and Orientation Search

We first tested bullet-cluster models with $P = 0.61, 0.45$, and 0.31 , while keeping $Q = 0.68$. This allowed us to study how the intrinsic shape of the bullet affected the projected dark matter contours. The $P = 0.31$ model produced a more elongated distribution and was selected for the orientation search.

We then tested different orientations of the $P = 0.31$ bullet, followed by small changes to the main-cluster orientation. Figure 6 shows the progression of this search. The first panel uses the geometry adopted by Lage & Farrar (2014), the second uses our $P = 0.31$ bullet, and the third shows the best match from the final parameter search. All three simulations are shown shortly after the first core passage, at a projected dark matter peak separation close to 750 kpc.

4. RESULTS AND DISCUSSION

4.1. X-ray Observations

The simulation results were compared with Chandra observations of the Bullet Cluster. The observations were processed using CIAO (Fruscione et al. 2026) and combined to obtain a total exposure time of 558 ks. An X-ray flux map was produced in the 0.5–2.0 keV energy band. The projected dark matter distribution was taken from the gravitational-lensing reconstruction of Rihtaršič et al. (2026).

The observed system has two main lensing mass peaks separated by approximately 750 kpc. The X-ray emission also contains two unequal components, with the main component being brighter and more extended. The gas has an asymmetric, wedge-shaped morphology, together with a sharp edge associated with the bow shock. These features were used to compare the observed Bullet Cluster with the simulations.

Figure 7 shows the Chandra X-ray flux map together with the gravitational-lensing mass contours. The simulated X-ray maps were produced in the same 0.5–2.0 keV energy band and using the same exposure time. The simulated snapshots were selected shortly after first core passage, when the projected dark matter peak separation was close to the observed value of 750 kpc.

4.2. Comparison with the Spherical Models

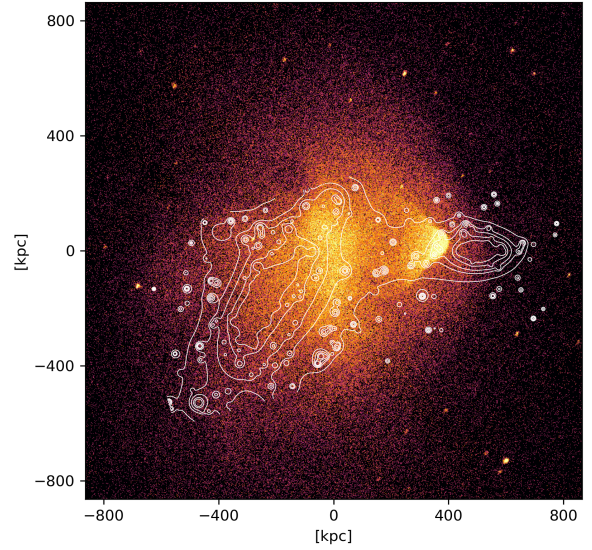


Figure 7. Observed Bullet Cluster X-ray flux map. The white contours show the projected mass distribution from the gravitational-lensing reconstruction of Rihtaršič et al. (2026).

Figure 2 shows the main effects of changing the initial velocity and impact parameter. In the upper row, the 250 km s^{-1} collision produces a broad gas distribution, while the 3000 km s^{-1} model produces a brighter and more compact central X-ray component. The 1000 km s^{-1} model provides an intermediate case and produces a clearer asymmetric gas morphology.

The lower row of Figure 2 shows the effect of the impact parameter. The head-on collision with $b = 0 \text{ kpc}$ remains too symmetric around the collision axis, while the $b = 600 \text{ kpc}$ model produces too much asymmetry. The $b = 300 \text{ kpc}$ model provides the most balanced result. These comparisons motivated the preferred combination of $v_i = 1000 \text{ km s}^{-1}$ and $b = 300 \text{ kpc}$.

The largest improvement to the spherical X-ray morphology resulted from changing the gas core radii. Figure 3 shows that increasing the main-cluster core radius produces a broader and more diffuse main gas component. A smaller bullet-cluster core radius produces a more compact and wedge-shaped X-ray peak, while a larger bullet core produces a broader and less defined secondary component.

The preferred spherical model used $r_{c,\text{main}}/r_{s,\text{main}} = 1/2$ and $r_{c,\text{bullet}}/r_{s,\text{bullet}} = 1/5$. This combination produced a diffuse main gas cloud together with a compact, wedge-shaped bullet component. It therefore provided a closer morphological match than the models with equal gas core radii. The dark matter contours remain similar throughout the grid because only the initial gas profiles were changed.

The spherical models reproduced the general separation of the gas and dark matter and part of the observed X-ray

morphology. However, their projected dark matter distributions remained more circular than the elongated mass components inferred from gravitational lensing. This difference motivated the use of triaxial initial conditions.

4.3. Comparison with the Triaxial Models

The triaxial search was primarily guided by the projected shapes and orientations of the gravitational-lensing mass contours. We began with the cluster geometry used by [Lage & Farrar \(2014\)](#) and then tested bullet-cluster models with $P = 0.61, 0.45$, and 0.31 , while keeping $Q = 0.68$. The $P = 0.31$ model produced the most elongated projected bullet and was therefore selected for the orientation search. This shape is more consistent with the recent lensing reconstruction, in which both mass components appear elliptical rather than circular.

Different bullet orientations were tested first, followed by smaller changes to the orientation of the main cluster. The progression of this search is shown in Figure 6. The preferred model produced a more elongated bullet component and improved the relative orientation of the two projected dark matter distributions. All models are shown shortly after first core passage, at the snapshot closest to a projected dark matter peak separation of approximately 750 kpc.

The triaxial simulations retained the preferred spherical gas-core configuration, with $r_{c,\text{main}}/r_{s,\text{main}} = 1/2$ and $r_{c,\text{bullet}}/r_{s,\text{bullet}} = 1/5$. The preferred model reproduced the broad wedge-shaped X-ray distribution while providing a better match to the observed dark matter contours than the spherical models. Nevertheless, the detailed X-ray structure and the sharpness of the observed bow shock were not fully reproduced.

4.4. Limitations and Future Work

The comparison in this work focused on the projected dark matter separation and the morphology of the dark matter and X-ray distributions. The snapshots were selected shortly after first core passage, when the dark matter peak separation was close to the observed value of 750 kpc. The preferred models were then identified by comparing the overall X-ray

morphology and projected dark matter contours with the observations.

The simulations used idealized adiabatic SPH and did not include magnetic fields, radiative cooling, star formation, feedback, or non-thermal pressure support. These effects may influence the detailed gas morphology and the sharpness of the bow shock. In addition, the large number of possible triaxial shapes and orientations limited the parameter space that could be explored.

Future work could include magnetohydrodynamical simulations and a more systematic search of the cluster shapes, orientations, impact parameters, and initial velocities. This would allow the parameter space to be constrained more efficiently and could provide a closer match to both the Chandra and gravitational-lensing observations.

5. CONCLUSIONS

In this work, we used idealized N -body/SPH simulations to model the Bullet Cluster using recent mass estimates from gravitational-lensing observations. The spherical simulations were used to study the effects of the initial velocity, impact parameter, and gas core radii. An initial velocity of 1000 km s^{-1} and an impact parameter of 300 kpc produced the preferred collision geometry. The closest X-ray morphology was obtained using a larger gas core in the main cluster and a more compact gas core in the bullet. Triaxial models were then used to better reproduce the elongated dark matter distributions seen in the lensing maps. These models improved the projected dark matter morphology while maintaining a peak separation close to the observed value of approximately 750 kpc.

The simulations reproduce the main features of the system, including the separation of the gas and dark matter and the broad shape of the X-ray-emitting gas. However, the observed X-ray morphology and shock structure were not fully reproduced. Future work could include magnetohydrodynamics and additional gas physics, as well as a more systematic search of the cluster shapes and orientations. These improvements would allow the observed dark matter and X-ray distributions to be compared with the simulations in greater detail.

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